FIN3080 Investment Analysis and Portfolio Management
Sample Exam

Summer 2018

CUHK(Shenzhen)

1. (20)Answer the following questions briefly and concisely:
   (1) What is the main conclusion of CAPM? Why is CAPM so important?

   (2) What are the factors in the Famma-French three factor model? What do these factors represent?

2. (20) Give the definitions of the following terms:
   (1) Market portfolio

   (2) Sharpe Ratio

   (3) Alpha

   (4) Risk factor

3. (10) Suppose that the riskless rate of return is 4% and the expected market return is 12%. The standard deviation of the market return is 11%. Suppose as well that the covariance of the return on Stock A with the market return is 1.65%.
   (a) What is the beta of Stock A?

   (b) What is the expected return on Stock A?

   (c) If the variance of the return on Stock A is 2.4%, what percentage of this variance is due to

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market risk?

4. (10) Consider a two-factor APT pricing framework. Suppose we have two assets A and B, with expected rate of return 16% and 26%, respectively. With respect to the two factors: for asset A, the factor loadings are 1.2 and 0.4, respectively; and for asset B, the factor loadings are 0.8 and 1.6, respectively. Suppose the risk-free rate is 6%. What are the risk premia for the two factors?

5. (15) Answer by "True" or "False" and provide a short explanation for your answer.
   (a) The CAPM is a special case of APT.

   (b) According to the CAPM, if $\alpha > 0$, then the security is overpriced.

   (c) Stock returns at 5-minute, daily, weekly, and monthly frequency are equally serially uncorrelated.

6. (25 points) In the recent finance literature it is suggested that asset prices are fairly well described by a so-called factor model, where excess returns are linearly explained from excess returns on a number of "factor portfolios". As in the CAPM, the intercept term should be zero, just like the coefficient for any other variable included in the model the value of which is known in advance (e.g. a January dummy). The data set contains excess returns on four factor portfolios for January 1960 to December 2002:
   - `rmrf`: excess return on a value-weighted market proxy
   - `smb`: return on a small-stock portfolio minus the return on a large-stock portfolio (Small minus Big)
   - `hml`: return on a value-stock portfolio minus the return on a growth-stock portfolio (High minus Low)
   - `umd`: return on a high prior return portfolio minus the return on a low prior return portfolio (Up minus Down)

   All data are for the USA. Each of the last three variables denotes the difference in returns on two hypothetical portfolios of stocks. These portfolios are re-formed each month on the basis of the most recent available information on firm size, book-to-market value of equity and historical returns, respectively. The *hml* factor is based on the ratio of book value to market value of equity, and reflects the difference in returns between a portfolio of stocks with a high book-to-market ratio (value stocks) and a portfolio of stocks with a low book-to-market ratio (growth stocks). The factors are motivated by empirically found anomalies of the CAPM (for example, small firms appear to have higher returns than large ones, even after the CAPM risk correction). In addition to the excess returns on these four factors, we have observations

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on the returns on ten different "assets" which are ten portfolios of stocks, maintained by the Center for Research in Security Prices (CRSP). These portfolios are size-based, which means that portfolio 1 contains the 10% smallest firms listed at the New York Stock Exchange and portfolio 10 contains the 10% largest firms that are listed. Excess returns (in excess of the risk-free rate) on these portfolios are denoted by r1 to r10, respectively. In answering the following questions, we use r1, r3, r7, and r10. We first estimate the following models:

```
> assets<-read.table("C:/data/assets2.dat", header=T)
> str(assets)
'data.frame':   528 obs. of  15 variables:
 $ RMRF: num  -6.99 0.99 -1.46 -1.7 3.08 2.09 -2.23 2.85 -6 -0.7 ...
 $ SMB : num  2.13 0.71 -0.65 0.32 1.42 -0.24 -0.4 0.34 -0.98 -3.75 ...
 $ HML : num  2.65 -2.15 -2.69 -2.22 -3.83 -0.3 1.94 0.67 1.29 2.34 ...
 $ UMD : num  -3.28 3.59 1.85 2.6 4.77 1.03 -0.8 2.04 0.3 1.27 ...
 $ RF  : num  0.33 0.29 0.35 0.19 0.27 0.24 0.13 0.17 0.16 0.22 ...
 $ R1  : num  0.3 -1.25 -1.12 -3.06 0.83 3.33 -1.47 5.4 -6.13 -3.17 ...
 $ R2  : num  -2.41 0.96 -3.56 -2.07 0.43 1.77 -1.23 3.95 -5.25 -2.36 ...
 $ R3  : num  -3.24 1.91 -3.08 -2.07 3.23 1.75 -2.51 2.66 -6.15 -3.69 ...
 $ R4  : num  -4.57 1.5 -2.83 -1.98 3.54 2.15 -1.56 4.1 -6.57 -1.81 ...
 $ R5  : num  -3.73 1.02 -2.69 -2.39 2.46 2.02 -1.7 2.97 -6.01 -3.81 ...
 $ R6  : num  -5.24 0.3 -2.72 -2.12 3.32 2.89 -1.59 4.02 -5.63 -2.05 ...
 $ R7  : num  -5.14 1.29 -1.92 -1.27 2.41 2.3 -2.4 2.94 -6.13 -1.85 ...
 $ R8  : num  -5.16 1.94 -1.94 -1.02 2.12 2.65 -0.77 2.81 -4.91 -1.27 ...
 $ R9  : num  -4.83 1.25 -0.58 0.07 3.33 2.75 -2.05 3.27 -5.79 -1.08 ...
 $ R10 : num  -7.61 1.25 -0.88 -1.89 3.28 2.31 -2.23 2.86 -5.86 0.09 ...
> capm_1<-lm(R1~RMRF, data=assets)
> summary(capm_1)

Call:
lm(formula = R1 ~ RMRF, data = assets)

Residuals:
     Min       1Q   Median       3Q      Max
-16.3405  -2.5297  -0.1227   1.8902  25.2605

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.76091    0.18100   4.204 3.08e-05 ***
RMRF         1.07919    0.04028  26.790  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.137 on 526 degrees of freedom
Multiple R-squared: 0.5771,     Adjusted R-squared: 0.5763
F-statistic: 717.7 on 1 and 526 DF,  p-value: < 2.2e-16

> capm_2<-lm(R3~RMRF, data=assets)
> summary(capm_2)

Call:
```


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```
lm(formula = R3 ~ RMRF, data = assets)

Residuals:
     Min       1Q   Median       3Q      Max
-12.3073  -1.6787  -0.1498   1.5627  14.2148

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.65974    0.12558   5.253 2.17e-07 ***
RMRF         1.17226    0.02795  41.942  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.871 on 526 degrees of freedom
Multiple R-squared: 0.7698,     Adjusted R-squared: 0.7694
F-statistic:  1759 on 1 and 526 DF,  p-value: < 2.2e-16

> capm_3<-lm(R7~RMRF, data=assets)
> summary(capm_3)

Call:
lm(formula = R7 ~ RMRF, data = assets)

Residuals:
    Min      1Q  Median      3Q     Max
-5.3427 -0.9043 -0.1046  0.8887  9.6437

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.60549    0.06852   8.836   <2e-16 ***
RMRF         1.09089    0.01525  71.531   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.566 on 526 degrees of freedom
Multiple R-squared: 0.9068,     Adjusted R-squared: 0.9066
F-statistic:  5117 on 1 and 526 DF,  p-value: < 2.2e-16
```

(a) What is the economic meaning of the estimated intercept?

(b) According to the regression results, does CAPM hold? Why?

Now we estimate the following four-factor model:

$$r_{jt} = \alpha_j + \beta_{j1} rmrf_t + \beta_{j2} smb_t + \beta_{j3} hml_t + \beta_{j4} umd_t + u_{jt},$$

```
> ffactor_1<-lm(R1~RMRF+SMB+HML+UMD, data=assets)
> summary(ffactor_1)
```


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```
Call:
lm(formula = R1 ~ RMRF + SMB + HML + UMD, data = assets)

Residuals:
    Min      1Q  Median      3Q     Max
-7.7633 -1.0827 -0.1227  0.9479  8.4539

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.455826   0.081085   5.622 3.08e-08 ***
RMRF        0.892562   0.019307  46.231  < 2e-16 ***
SMB         1.249266   0.025408  49.169  < 2e-16 ***
HML         0.271603   0.029762   9.126  < 2e-16 ***
UMD         0.003524   0.019462   0.181    0.856
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.75 on 523 degrees of freedom
Multiple R-squared: 0.9248,     Adjusted R-squared: 0.9242
F-statistic:  1608 on 4 and 523 DF,  p-value: < 2.2e-16

> ffactor_2<-lm(R3~RMRF+SMB+HML+UMD, data=assets)
> summary(ffactor_2)

Call:
lm(formula = R3 ~ RMRF + SMB + HML + UMD, data = assets)

Residuals:
    Min      1Q  Median      3Q     Max
-3.2669 -0.5843 -0.0041  0.5703  4.5296

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.47486    0.04386  10.827   <2e-16 ***
RMRF         1.03364    0.01044  98.981   <2e-16 ***
SMB          0.90135    0.01374  65.587   <2e-16 ***
HML          0.19049    0.01610  11.833   <2e-16 ***
UMD         -0.03303    0.01053  -3.138   0.0018 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.9464 on 523 degrees of freedom
Multiple R-squared: 0.9751,     Adjusted R-squared: 0.9749
F-statistic:  5125 on 4 and 523 DF,  p-value: < 2.2e-16

> ffactor_3<-lm(R7~RMRF+SMB+HML+UMD, data=assets)
> summary(ffactor_3)

Call:
```


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```
lm(formula = R7 ~ RMRF + SMB + HML + UMD, data = assets)

Residuals:
    Min      1Q  Median      3Q     Max
-4.6531 -0.6695 -0.0425  0.6271  5.1980

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.483987   0.051368   9.422  < 2e-16 ***
RMRF        1.050841   0.012231  85.917  < 2e-16 ***
SMB         0.366254   0.016096  22.754  < 2e-16 ***
HML         0.133521   0.018854   7.082 4.62e-12 ***
UMD         0.003477   0.012329   0.282    0.778
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.109 on 523 degrees of freedom
Multiple R-squared: 0.9536,     Adjusted R-squared: 0.9532
F-statistic:  2686 on 4 and 523 DF,  p-value: < 2.2e-16
```

(c) Does the above regression support the four-factor model? Why?

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